Lecture 9: An Overview of Fiscal Policy

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Outline

- General objective of government policies
- What is fiscal policy?
- Fiscal policy in historical perspective
- The government budget constraint
  - The intertemporal government budget constraint
  - How much can a government borrow?
  - Do government deficits crowd out investment?
- Fiscal problems in Europe
- The fiscal problem of the 21st Century
- Fiscal rules across countries (optional)

General Objective of Government Policies

- Objective of government policies
  - maintain full employment
  - moderate business cycle
  - achieve sustained long-term growth
  - keep price stability
- Duties of the treasurer of the Commonwealth of Australia: "The Treasurer has overall responsibility for all of the policy matters falling within the Treasury portfolio. These include: economic and fiscal policy; taxation; price surveillance; competition policy; national and occupational superannuation; borrowing money on the public credit of the Commonwealth; banking; insurance; currency and legal tender; foreign exchange; foreign investment in Australia; census and statistics; business law and practice; corporate and securities law; corporate insolvency; valuation services; and Commonwealth-State financial relations."

What is Fiscal Policy? (1 of 9)

- Fiscal policy is made by the president (premier, prime minister) and the Congress.
- It is the use of government budget (**expenditure**, **taxation**, **debts**) to achieve macroeconomic objectives.
- The two main instruments of fiscal policy are government expenditure and taxation.
- The government finances its spending by taxing the private sector. It budget or fiscal balance equals tax revenue minus expenditure. Three states of budget balance:
  - balanced budget
  - budget deficit
  - budget surplus

What is Fiscal Policy? (2 of 9)

- Government/public debt is the total amount that the government has borrowed. It is the accumulation of all past deficits. Budget surpluses allow the government to pay off some debts.
- The law of motion for nominal debt

$$D_t = D_{t-1} - (B_t - i_t D_{t-1})$$

  where D_t denote the stock of government debt at the end of year t, let i_t be the (average) nominal interest rate and let B_t be the primary (i.e., non-interest) government balance.
- Ratio of public debt to GDP becomes

$$\frac{D_t}{P_t Y_t} = \frac{(1+i_t)D_{t-1} - B_t}{P_t Y_t} = \frac{(1+i_t)D_{t-1} P_{t-1} Y_{t-1}}{P_t Y_t \, P_{t-1} Y_{t-1}} - \frac{B_t}{P_t Y_t}$$

$$= \frac{1+i_t}{(1+g_t)(1+\pi_t)} \cdot \frac{D_{t-1}}{P_{t-1} Y_{t-1}} - \frac{B_t}{P_t Y_t}$$

  where g_t is the growth rate of real GDP, and π_t is the inflation rate (measured as the rate of change of the GDP deflator, P).

What is Fiscal Policy? (3 of 9)

- There are three possible stances of fiscal policy:
  - **Neutral stance:** A balanced budget where government spending is fully funded by tax revenue. The overall budget outcome has a neutral effect on the level of economic activity.
  - **Expansionary stance:** Involves a net increase in government spending through rises in government expenditure, falls in taxation revenue, or a combination of both. It leads to a smaller budget surplus or a larger budget deficit.
  - **Contractionary stance:** Occurs when net government spending is reduced either through higher taxation revenue, reduced government spending, or a combination of both. It leads to a smaller budget deficit or a budget surplus.

What is Fiscal Policy? (4 of 9)

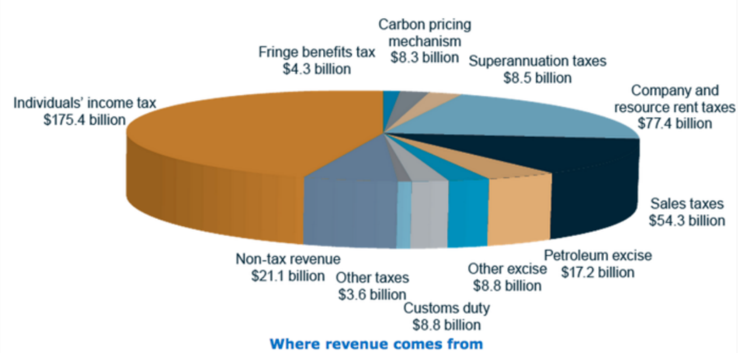
- Changes in the **level** and **composition** of taxation and government spending can impact the following variables in the economy:
  - Aggregate demand and the level of economic activity
  - The pattern of resource allocation
  - The distribution of income

Figure 1: Australian government budget aggregate

	Actual	Estimates			Projections	
	2011-12 $b	2012-13 $b	2013-14 $b	2014-15 $b	2015-16 $b	2016-17 $b
Receipts	**329.9**	**350.4**	**376.0**	**401.2**	**428.9**	**453.6**
Per cent of GDP	22.4	23.0	23.5	23.9	24.3	24.4
Payments(a)	**371.0**	**367.3**	**391.2**	**409.1**	**425.0**	**443.7**
Per cent of GDP	25.2	24.2	24.5	24.4	24.0	23.8
Net Future Fund earnings	2.2	2.5	2.8	2.9	3.1	3.3
Underlying cash balance(b)	**-43.4**	**-19.4**	**-18.0**	**-10.9**	**0.8**	**6.6**
Per cent of GDP	-2.9	-1.3	-1.1	-0.6	0.0	0.4
Revenue	**338.1**	**360.0**	**387.7**	**411.6**	**438.1**	**468.6**
Per cent of GDP	22.9	23.7	24.3	24.5	24.8	25.2
Expenses	**377.7**	**381.4**	**398.3**	**415.7**	**431.0**	**454.7**
Per cent of GDP	25.6	25.1	24.9	24.8	24.4	24.4
Net operating balance	-39.6	-21.5	-10.6	-4.1	7.1	13.8
Net capital investment	4.9	-1.2	2.9	2.2	1.1	3.0
Fiscal balance	**-44.5**	**-20.3**	**-13.5**	**-6.3**	**6.0**	**10.8**
Per cent of GDP	-3.0	-1.3	-0.8	-0.4	0.3	0.6
Memorandum item:						
Headline cash balance	-47.0	-22.2	-25.3	-21.0	-8.6	-3.3

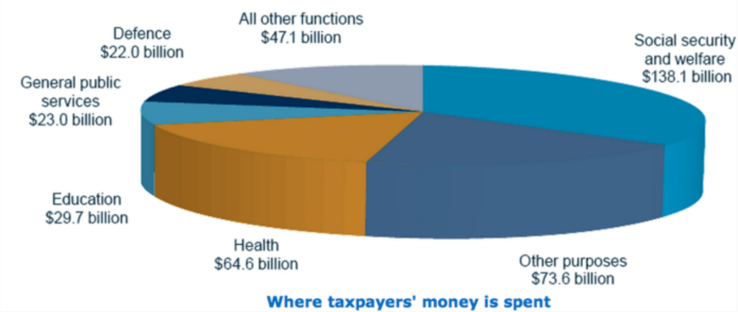
source: `http://www.budget.gov.au/2013-14/content/overview/html/overview_38.htm`

Figure 2: Australian government budget: revenue outcome



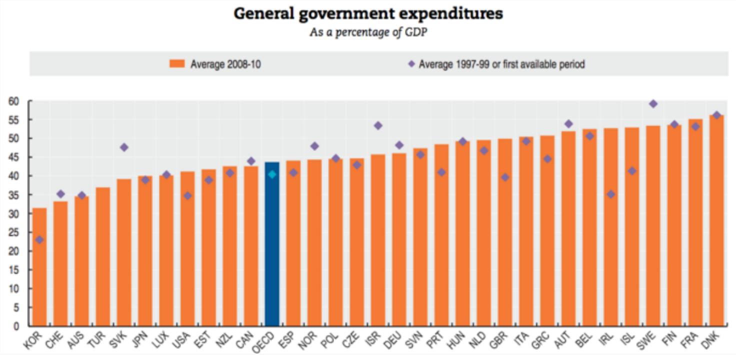
source: `http://www.budget.gov.au/2013-14/content/overview/html/overview_44.htm`

Figure 3: Australian government budget: expenses outcome



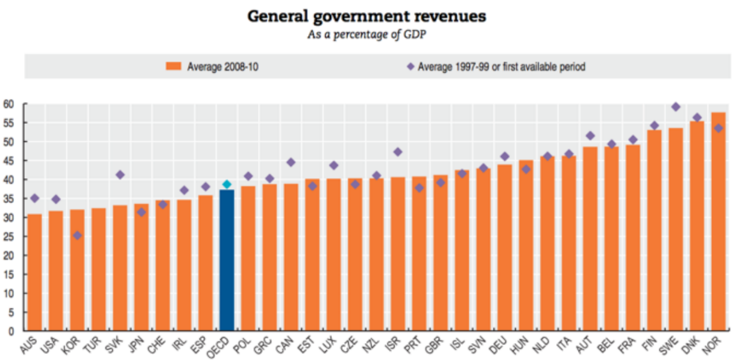
source: `http://www.budget.gov.au/2013-14/content/overview/html/overview_44.htm`

Figure 4: Government spending of OECD countries



source: `http://www.oecd-ilibrary.org/`

Figure 5: Government revenue of OECD countries



source: http://www.oecd-ilibrary.org/

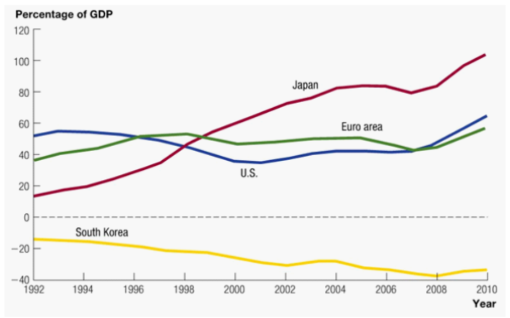
- The idea of using fiscal policy to combat recessions was introduced by John Maynard Keynes in the 1930s, partly as a response to the Great Depression.
- From the end of the Great Depression **until the early 1970s**, Keynesian economics provided the main inspiration for economic policymakers in industrialized countries. A main theme of Keynesian economics is the use of active fiscal policies to manage aggregate demand.
- The influence of Keynes's theories waned in the 1970s, due to stagflation and critiques from Friedman, Lucas, and other economists. Indirect intervention through monetary policy became more popular among industrialized countries.

Fiscal Policy in Historical Perspective (2 of 8)

- **Several concerns:**
- Time lags: The legislative process can be too slow to permit the effective use of fiscal policy when it is needed.
- The effects of tax and spending policies on economic activities are direct and strong, such that fiscal policies may distort economic activities and create extra fluctuations (e.g., the crowding-out effect of government spending).

- Concerns that active fiscal policy may lead to large and persistent budget deficits:
  - **Long-term budgetary challenge:** There is likely to be a sharp rise in the number of retirees relative to the number of workers in coming decades, such that the resulting increase in social security and health care spending may further raise the deficit.
  - There is widespread perception that large and persistent budget deficits could slow down economic growth and may lead to a debt crisis.
- The global financial crisis in 2007 caused a resurgence in Keynesian thought. Many governments implemented fiscal stimulus packages to combat recession.

Figure 6: The evolution of government debt to GDP around the world (net)



source: Charles Jones (website)

- Debt to GDP ratio for South Korea is negative.
- Norway is another country in OECD with substantial negative debt-GDP ratio (petroleum and natural gas reserve).

Figure 7: Evolution of government debt to GDP (per cent of potential GDP)

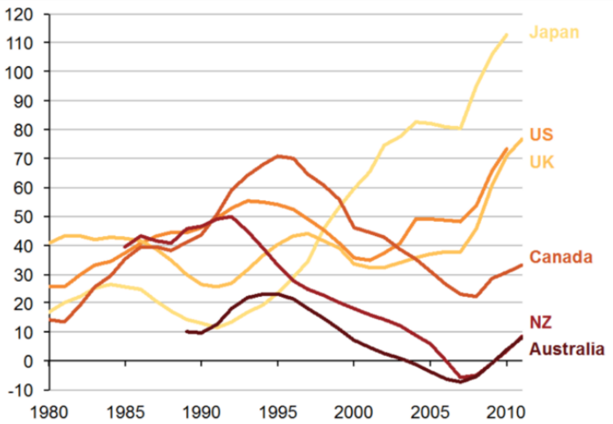
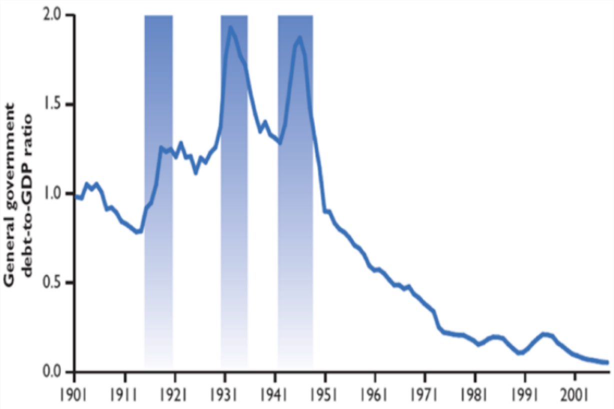
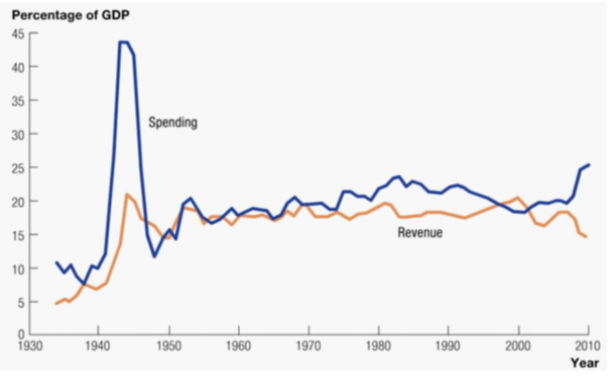


Figure 8: Evolution of government debt to GDP, Australia 1901-2005



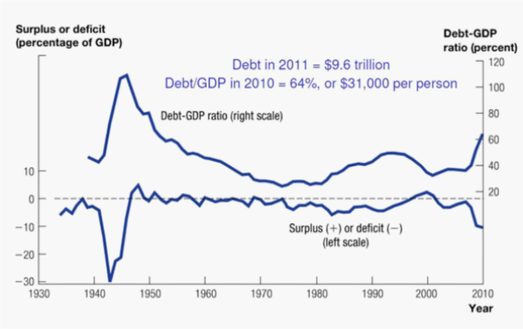
source: Blanchard and Sheen (2009)

Figure 9: Federal revenue and spending since 1930 in the U.S.



Tax revenues were very small in the 1930s.

Figure 10: Federal debt and deficits in the U.S.



- When the government runs a deficit, it must borrow, by issuing government bonds, to finance its spending.
- The outstanding stock of bonds that have been issued in the past is called government debt.

The Government Budget Constraint (1 of 4)

The intertemporal government budget constraint:

- How much can the federal government borrow, and how does borrowing affect the economy?
- The government budget constraint (**flow version**), based on an accounting identity:

$$G_t + Tr_t + iD_t = T_t + \Delta D_t + \Delta M_t$$

The Government Budget Constraint (2 of 4)

- **Sources of government expenses:**
  - G_t: spending on goods and services
  - Tr_t: transfer payments (unemployment insurance, social security, welfare programs)
  - D_t: existing stock of government debt
  - i: nominal interest rate
- **Sources of government funds:**
  - T_t: tax revenue
  - ΔD_t: new borrowing
  - ΔM_t: change in the stock of money, introducing interaction between monetary and fiscal policy

- Assuming $\Delta M_t = 0$ and $Tr_t = 0$:

$$D_{t+1} = (1+i)D_t + (G_t - T_t)$$

- Suppose an **economy exists for two periods**: The economy begins in period 1 and ends after period 2.
- The present value of spending plus the initial debt should equal the present value of taxes:

$$G_1 + \frac{G_2}{1+i} + (1+i)D_1 = T_1 + \frac{T_2}{1+i}$$

The Government Budget Constraint (4 of 4)

- Or, period-1 balance plus the present value of period-2 balance equals initial debt:

$$T_1 - G_1 + \frac{T_2 - G_2}{1 + i} = (1 + i)D_1$$

- The Lesson of Budget Constraints: Budgets must balance NOT period by period, but in present discounted value.

How much can a government borrow? (1 of 3)

- Suppose the outstanding debt is growing at 3% per year: is that a problem?
- Is there a particular debt/GDP ratio at which problems occur?
- What kind of problems happen?
- Is borrowing from future generations a good thing?
- Do government deficits crowd out investment?

How much can a government borrow? (2 of 3)

- Is there a particular debt/GDP ratio at which problems occur?
  - Argentina 2001: Debt/GDP peaked at 65%.
  - Greece: About 170% in 2012.
  - But Japan today and the U.S. after WWII were higher.
- Is borrowing from future generations a good thing?
  - Under current policies, the generation born in 2004 will pay a net lifetime tax rate of 18.5%.
  - On average, future generations will have to pay a net lifetime tax rate of 58.2%.
  - These numbers are from Gokhale (2008), based on Auerbach et al. (1994).

How much can a government borrow? (3 of 3)

- If GDP is growing faster than debt, then the ratio of debt to GDP will decline.
- Probability of default:
  - When the debt–GDP ratio becomes too high, lenders may worry about the government's ability to repay and may stop lending.
  - If investors doubt the government's ability to finance its debt, they may demand a higher interest rate on new borrowing.
- Inflation:
  - Repaying debt with a lower real value.

Do government deficits crowd out investment? (1 of 2)

- Recall our manipulation of the National Income Identity:

$$(Y - T - C) + (T - G) + (IM - EX) = I$$

- Potential offsetting forces:
  - Ricardian equivalence (private saving may rise)
  - Foreign saving may increase to take advantage of a high return

Do government deficits crowd out investment? (2 of 2)

Figure 11: U.S. investment and the budget deficit

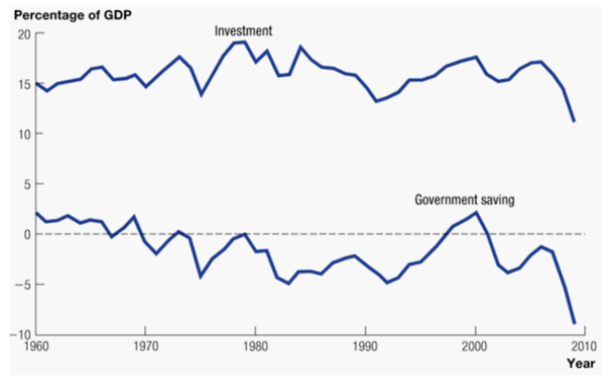
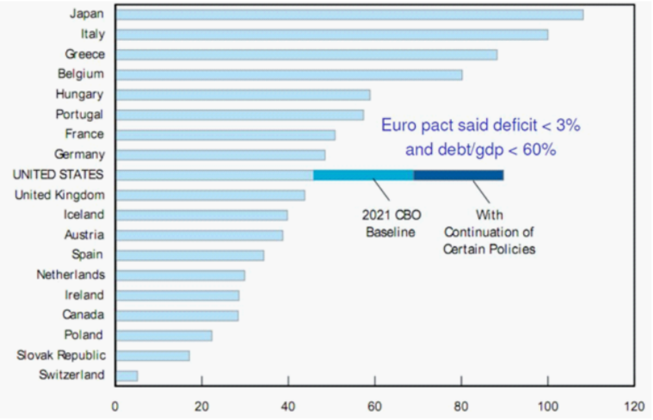


Figure 12: Debt burdens around the OECD in 2009



Source: Economist.com

The reckoning

% of GDP (unless stated)

	Gross government debt 2010 est	Gross government debt 2015 f'cast	Primary budget adjustment, 2010-15 f'cast	Net international-investment position, 2009	% of gov't bonds held abroad*	Ten-year gov't-bond yield†, %
Greece	140.2	165	10	-89.1	58.0	11.3
Ireland	97.4	125	13	-102.5	54.2	8.3
Portugal	82.8	100	8	-113.2	66.0	6.8
Spain	64.4	85	10	-95.7	38.7	5.5

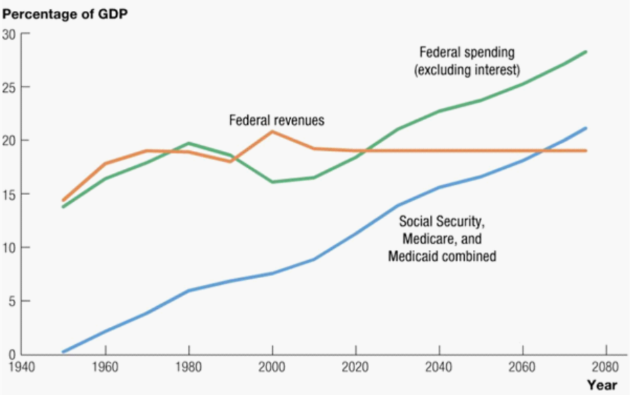
Sources: *The Economist*; European Commission; IMF; Citigroup; Thomson Reuters *Q3 2010 †January 12th 2011

Source: Economist.com

Fiscal problems in Europe (4 of 4)

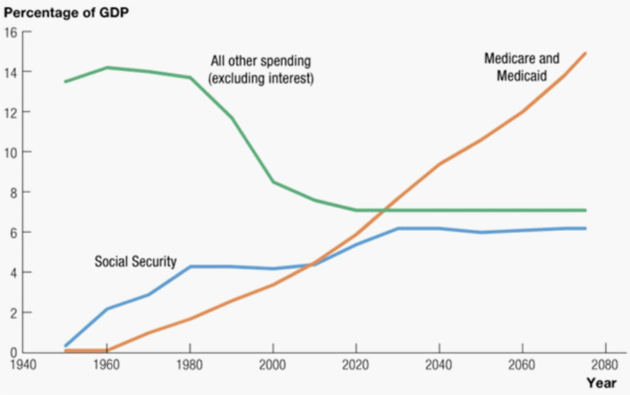
- Issues in the Euro Debt Crisis
- Austerity
  - Goal is to reduce debt and deficits.
  - By prolonging the recession, does it perversely raise them?
- Long-term fiscal balance
  - Reform entitlement programs and spending.
  - Raise retirement ages, limit pensions, reform labour market institutions.
  - If phased in, maybe solves fiscal problems, promotes long-run growth, and mitigates short-run problems?

Figure 13: Federal spending and revenue, 1950-2075



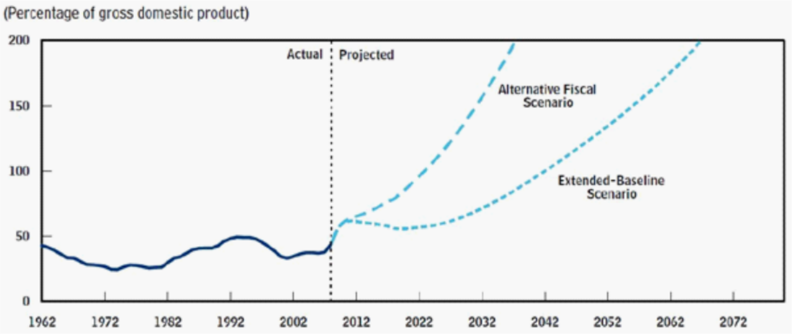
Source: CBO 2002: "A 125-Year Picture of the Federal Government..."

Figure 14: Components of Federal spending, 1950-2075



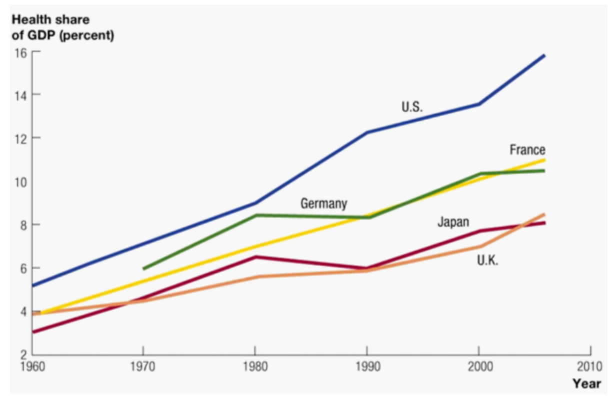
Source: CBO 2002: "A 125-Year Picture of the Federal Government..."

Figure 15: The Debt-GDP ratio in the future?



Source: Charles Jones (website)

Figure 16: Health spending as a share of GDP around the world



Source: Charles Jones (website)

Why is health spending as a share of GDP growing?

- Joseph Newhouse (1992): New, expensive technologies.
- But why are these being invented and used?
- One possible explanation (Hall and Jones, 2007):
  - Standard economics predicts that the health share should rise.
  - Diminishing returns to consumption.
  - As we get richer, the extra utility from another car, another television, or more clothing declines — there is barely enough time to enjoy what we have.
  - Instead, what we value more are additional days of life to enjoy higher incomes.
  - It may be optimal for health shares to continue rising.

How to finance higher health spending?

- Most health spending occurs toward the end of life.
  - Can we look forward and be disciplined enough to save 30% of our lifetime income for spending that may occur in 30 to 50 years?
- Who provides insurance?
  - Classic result in economics: private insurance often breaks down; private insurers have large incentives to avoid paying out.
  - But tax distortions from government provision could be enormous.
  - Health savings accounts? Required minimums; optional extra contributions.
  - See David Cutler, *Your Money or Your Life: Strong Medicine for America's Health Care System*, 2005.

Appendix: Fiscal Rules (1 of)

- During the 1970s and 80s many industrialized countries had large public-sector budget deficits.
- Many countries recognize this as a serious issue, and gave adopted some sort of fiscal rules that aims to limit the size of budget deficit.
- Debate in the U.S. on balanced budget rule
  - A Balanced-Budget Amendment proposal was narrowly rejected in 1995, but is still pursed by the Congress.
  - Argument against the proposal: It would eliminate the use of fiscal policy as macroeconomic policy instrument.
  - Argument in favour of the proposal: The usefulness of macroeconomic policy, the fiscal policy in particular, is limited (time lag, side effect).

Fiscal rules of the Euro area

- Main requirement: an annual budget deficit no higher than 3% of GDP; a national debt lower than 60% of GDP or approaching that value.
- These numerical rules were first stated in the Maastricht Treaty signed in 1991 as a criteria for countries to join the EMU.
- The Treaty also allows for flexibility in the implementation of the criteria, since it would be impossible for some countries to comply with the fiscal rules before the introduction of the EMU in 1999.
- In 1997, the stability of Growth pact (SGP) was signed to ensure that EMU countries continue to observe Maastricht criteria once they join the EMU.

Timeline of the Euro area

- 1998: European Central Bank created
- 1999: Introduction of the euro. Belgium, Germany, Ireland, Spain, France, Italy, Luxembourg, the Netherlands, Austria, Portugal, and Finland adopt the euro.
- 2001: Greece adopts the euro.
- 2007: Slovenia adopts the euro.
- 2008: Cyprus and Malta adopt the euro.
- Euro area.

Fiscal policy constraints in Australia

- In Australia, no-one is pushing for a balanced-budget amendment.
- However the coalition government from 1996-2007 built a political platform based on the virtue of budget-surpluses. For electoral reasons only, Treasurer Costello created this artificial constraint on fiscal policy.
- In 2008-09, all countries have become more flexible in their fiscal policies as a necessary response to the global financial crisis that has threatened to bankrupt many financial institutions and push the economy towards recession.